

MICHAEL OKAFOR

Full-Stack & ML Engineer · Python/FastAPI · TypeScript/NestJS/Next.js · Ruby on Rails

Fredericton, Canada · Remote · mikuli2015@gmail.com · [GitHub](#) · [LinkedIn](#) · [Portfolio](#)

PROFESSIONAL SUMMARY

Software engineer with 10 years of experience across Python, Ruby on Rails, and TypeScript/Node.js, preceded by six years in business analysis within retail/commercial banking. Founder of ElderCare Companion, a multi-tenant home care platform for Atlantic Canada, built on a Python/FastAPI core with a companion TypeScript/NestJS compliance-training platform — PIPEDA/PHIPAA-compliant, bilingual EN/FR, and covered by 3,000+ automated tests. Background includes directing a monolith-to-API-first re-architecture that cut latency by 84%, leading and mentoring Agile teams, university-level teaching in programming and data structures, and four peer-reviewed publications spanning applied AI and web-based information systems. Uses AI coding tools (Claude) deliberately for velocity — decomposing work into specs, directing implementation, and gating every change behind automated tests — while retaining full ownership and review of everything shipped.

CORE TECHNICAL SKILLS

Languages & Frameworks: Ruby on Rails, Python (FastAPI, Flask), TypeScript, JavaScript, React, React Native, Next.js, NestJS

Backend & Data: PostgreSQL, Redis, SQLAlchemy, Prisma, RESTful APIs, GraphQL, WebSockets, multi-tenant SaaS architecture

ML & AI: scikit-learn, TensorFlow, RAG pipelines, AWS SageMaker

Compliance & Security: PIPEDA & PHIPAA compliance, MFA/TOTP, JWT auth, RBAC, audit logging

Cloud & DevOps: AWS, Docker, GitHub Actions, DigitalOcean, Vercel, CI/CD

Testing & Quality: TDD, pytest, vitest, Playwright (E2E + accessibility), 3,000+ automated tests

AI-Assisted Development: Claude Code, Cursor — used daily as a core part of the workflow

PROFESSIONAL EXPERIENCE

Founder — ElderCare Companion — Flagship Project, Remote

Jan 2026 – Present

- Architected and built a production multi-tenant SaaS platform: Python/FastAPI core (227 API endpoints, GPS-verified shift check-in, digital MAR) plus ElderCare Academy, a separate TypeScript/NestJS/Next.js compliance-training platform delivered as a subscription entitlement.
- PIPEDA/PHIPAA-compliant, bilingual EN/FR, spanning five Atlantic jurisdictions; 3,000+ automated tests including Playwright E2E and accessibility coverage.

Software Engineer — Introhive Services Inc., Fredericton, Canada

Mar 2025 – Dec 2025

- Architected PostgreSQL solutions with GIN indexing, achieving sub-2ms query performance and a 32% reduction in API latency.
- Reduced post-release defects by 25% through automated regression and integration testing; executed zero-downtime Rails migrations.

Software Engineer — Themiscore, Fredericton, Canada

Sep 2024 – Mar 2025

- Built Rails-based RESTful APIs for document ingestion and semantic search; reduced FastAPI endpoint latency by 40%.

Lecturer & Software Engineer / Research & Data Developer — Nnamdi Azikiwe University, Awka, Nigeria

Dec 2019 – Dec 2023

- Taught undergraduate Programming Fundamentals, Numerical Computations, and Data Algorithms and Structures while engineering production applications serving 500+ users.

Senior Software Engineer / Lead — Lodestar Digital, Lagos, Nigeria

2016 – 2019

- Led a 6-person Agile team; directed a monolith-to-API-first re-architecture, reducing P95 latency from 1.8s to 280ms and cutting server costs by 44%.

Business Developer / Analyst — Diamond Bank plc, Lagos, Nigeria

2010 – 2016

- Gathered and analyzed business requirements from stakeholders across retail/commercial banking business units.

SELECTED PROJECTS

ElderCare Companion — Python, FastAPI, PostgreSQL [\[link\]](#)

- Multi-tenant home care management platform for Atlantic Canada — GPS-verified visits, digital MAR, PIPEDA/PHIPAA-compliant, bilingual EN/FR.

MapleMatch — *Next.js, FastAPI, PostgreSQL* [\[link\]](#)

- AI-powered housing-matching platform with rules + semantic + ML scoring, live across all 13 Canadian provinces; bilingual EN/FR.

ElderCare Academy (Maple Care LMS) — *Next.js, NestJS, Prisma* [\[link\]](#)

- Multi-tenant compliance-training SaaS with fail-closed tenant isolation and inspector-ready audit exports.

Predictive Sensor Analytics — *Python, FastAPI, PostgreSQL* [\[link\]](#)

- Anomaly detection for high-frequency manufacturing sensor streams using Isolation Forest, Autoencoder, and LSTM models.

RAG Legal Document Platform — *Ruby on Rails, React, PostgreSQL* [\[link\]](#)

- Retrieval-augmented generation platform for legal document search and AI-assisted summarization.

PUBLICATIONS

- "Agent Based Intelligent System for Enhanced Teamwork Performance" — International Journal on Data Science and Technology, May 2024. [\[Read Paper\]](#)
- "Semantic Web Ontology Technology and Its Impact on E-Learning" — American Journal of Embedded Systems and Applications, Nov 2021. [\[Read Paper\]](#)
- "Medical Diagnostic Expert System for Malaria and Typhoid Fever" — International Journal of Scientific Research and Development, 2022. [\[Read Paper\]](#)
- "Web Based Lands and Housing Information Tracking System" — ResearchGate, 2023. [\[Read Paper\]](#)

EDUCATION & ASSOCIATIONS

Post Graduate Diploma, Applied Artificial Intelligence and Machine Learning — Conestoga College, Canada, 2024

Master of Science, Information Technology — National Open University of Nigeria

Bachelor of Science, Computer Science — Nnamdi Azikiwe University, Awka, Nigeria

Member: Nigeria Computer Society · Computer Professionals of Nigeria · CIVITEC Fredericton, NB